

GCOMP

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2026-05-01

G-Comp With Two Regression Models

This document has been slightly edited by Mira Terdiman.

```
data <- read.csv("data/clean_state_data.csv")

#Georgia value added
data$phf_per_capita[data$state == "Georgia" & is.na(data$phf_per_capita)] <- 108

#Mississippi value added
Mi_phf <- (567831325/2954160)
data$phf_per_capita[data$state == "Mississippi" & is.na(data$phf_per_capita)] <- Mi_phf

data$phf_per_capita[is.na(data$phf_per_capita)] <- median(data$phf_per_capita, na.rm = TRUE)

names(data)

## [1] "state"           "measles_cases"   "population"
## [4] "cases_per_100k"  "median_hh_incomeE" "median_hh_incomeM"
## [7] "pct_insuredE"    "pct_insuredM"    "pct_bachelors_plusE"
## [10] "pct_bachelors_plusM" "pct_povertyE"    "pct_povertyM"
## [13] "pct_urban"       "pct_rural"       "phf_per_capita"
## [16] "policy"          "exposure_binary"

model1 <- lm(cases_per_100k ~ exposure_binary + pct_povertyE + pct_urban + phf_per_capita, data = data)
model2 <- lm(cases_per_100k ~ exposure_binary + pct_povertyE + pct_urban + phf_per_capita + pct_insuredE, data = data)

data_A1 <- data
data_A1$exposure_binary <- 1

data_A0 <- data
data_A0$exposure_binary <- 0

#G-comp with the parsed model (model1)
Y1_pred <- predict(model1, newdata = data_A1)
Y0_pred <- predict(model1, newdata = data_A0)

ATE <- mean(Y1_pred - Y0_pred)

#G-comp with the full model (model2)
Y1_pred2 <- predict(model2, newdata = data_A1)
Y0_pred2 <- predict(model2, newdata = data_A0)

ATE2 <- mean(Y1_pred2 - Y0_pred2)
```

```

#two ATEs
ATE

## [1] 0.4621942

ATE2

## [1] 0.4160642

#fairly similar

nobs(model1)

## [1] 51

# no states were dropped
used_rows <- as.numeric(rownames(model.frame(model1)))
used_states <- data$state[used_rows]

dropped_states <- setdiff(data$state, used_states)

dropped_states

## character(0)

```

Method: Fit two linear regression models estimating $E[Y \mid A, W]$

We kept the first model simple and focused on a few key confounders from our DAG, especially since our sample size is small, to help avoid overfitting. Adjusted for: Poverty, Urban population, Public health funding

Predicted outcomes under: $A = 1$ (more permissive policies) $A = 0$ (more restrictive policies)

Estimated ATE as the average difference in predicted outcomes

Result:

ATE = 0.462 additional measles cases per 100,000 or 3.0 additional measles cases per 10,000 people ATE2 = 0.416

Interpretation:

States with more permissive vaccine exemption policies had, on average, about 0.462 additional measles cases per 100,000 people or 4.62 additional measles cases per 10,000 people compared to states with stricter policies after adjustment. This suggests that more permissive policies may lead to higher measles incidence at the state level. However, these results should be interpreted cautiously due to the small sample size and the possibility of unmeasured confounding.

G-Comp With Super Learner

<https://ehsanx.github.io/TMLEworkshop/tmle.html> (using this as a reference)

```

set.seed(252)
#reformatting data for the superlearner
final_df <- data |>
  dplyr::select(cases_per_100k, exposure_binary, pct_urban, pct_insuredE, pct_bachelors_plusE, pct_povertyE,
  rename(Y = cases_per_100k,
    A = exposure_binary,
    W1 = pct_urban,
    W2 = pct_insuredE,
    W3 = pct_bachelors_plusE,

```

```

      W4 = pct_povertyE,
      W5 = phf_per_capita)

slib <- c("SL.mean", "SL.glm", "SL.bayesglm", "SL.earth", "SL.gam")

#modeling
model_sl <- SuperLearner(Y = final_df$Y,
                        X = final_df[, c("A", "W1", "W2", "W3", "W4", "W5")],
                        SL.library = slib,
                        family = "gaussian")

#estimating potential outcomes
data_a1 <- final_df[, c("A", "W1", "W2", "W3", "W4", "W5")]
data_a1$A <- 1
pred_y1 <- predict(model_sl, newdata = data_a1)$pred

data_a0 <- final_df[, c("A", "W1", "W2", "W3", "W4", "W5")]
data_a0$A <- 0
pred_y0 <- predict(model_sl, newdata = data_a0)$pred

ATE_SL <- mean(pred_y1 - pred_y0)

ATE_SL

```

```
## [1] 0.1653136
```

Much lower ATE than previously... interesting!